SECTION 00 31 32

Geotechnical Data Report

for

Fernandina Harbor

Maintenance Dredging

Nassau County, Florida

Prepared by

Geosystems Branch

Engineering Division

Jacksonville District Corps of Engineers

17 September 2024

SECTION 00 31 32

GEOTECHNICAL DATA REPORT

Table of Contents

List of Tables

List of Plates

SECTION 00 31 32

GEOTECHNICAL DATA REPORT

1 SCOPE

The information provided in this section encompasses the geotechnical field investigations available for this project. The investigations consist of vibracore borings and a surficial sample. The associated boring logs and laboratory data are presented in paragraphs 4.5 and 4.6, respectively. A character of materials paragraph is included to provide a comprehensive description of the materials utilizing both recent and historical knowledge of the project area. Also included in this section are definitions of terms and boring log notes, which provide additional explanation of the boring logs and drilling techniques.

Items discussed in the character of materials paragraph may not appear explicitly on the boring logs. Based on historic knowledge of the project area, the character of materials paragraph includes items that supplement the data documented by the boring logs. When reviewing the boring logs, use all of the data on the logs, including the materials description, legend, and blow counts. When evaluating the subsurface conditions, use all of the data, including the character of materials paragraph and boring logs.

During the solicitation period, any questions that pertain to the information provided in this section should be addressed to the contract specialist identified in Block 9 of SF1442. After contract award, questions should be addressed to the appointed ACO/COR, which will coordinate with Geosystems Branch.

2 CHARACTER OF MATERIALS

2.1 THE LOCAL GEOLOGY

The Fernandina Harbor is located along the St. Mary's River, which separates Cumberland Island to the north from Amelia Island to the south and is part of southeast Georgia and northeast Florida coastal plain. Surficial sediments composed of sands, shell, soft silts and clays are Holocene (present to 12k years Before Present (BP)) and Pleistocene (12k to 1.8 Million Years ago (Mya)) in age. These surficial sediments represent barrier island and marsh/lagoon facies of numerous shoreline complexes. Sparse Pliocene (1.8 Mya to 5.3 Mya) siliciclastics underlie these Pleistocene sediments. Extensive deposits of the Miocene (5.3 Mya to 23.8 Mya) aged Hawthorn Group lie just under most surficial sediments. The composition of the Hawthorn Group is highly variable, reflecting different depositional environments and repeated reworking of the sediment in response to sea level fluctuations on the shallow water shelf. The lithologies of the Hawthorn Group include limestone, dolomite, clay, and quartz sands with a notable quantity of phosphorite.

2.2 MATERIALS ENCOUNTERED

The materials encountered primarily consist of organic silts, silty sand, fine to medium sized quartz sand, sandy clay, and fat clay with varying amounts of sand sized to gravel sized shell and plant/ organic debris. Although not identified during any sampling or testing, some of the materials encountered consist of OH with organics and have the potential to produce noxious gas if in an anoxic environment. The materials to be removed from the maintenance areas accumulated as a result of shoaling since the previous dredging event in 2022.

3 DEFINITIONS

Definitions not explicitly indicated in the sections below are typical industry standard definitions from their respective ASTMs.

3.1 DEFINITIONS OF BASIC TERMS

Carbonate – Soil component that reacts with HCl of an indeterminate origin (shell, rock, etc.).
Fill – Material that has been placed by man, described with all soil characteristics.
Layer – Rock or soil with a thickness of 6 inches or less.
Lens – A geologic deposit of variable thickness, which disappears laterally in all directions and cannot be correlated to adjacent borings.
Rock – A naturally occurring substance composed of one or more minerals bound together. This geologic term includes a range of engineering properties: strength, hardness, permeability, weathering, and discontinuity. These properties are noted or can be inferred from the boring logs as blow counts, penetration rate, RQD, hardness, etc.
Shell – Material composed of predominantly (>75%) coarse-grained sand to gravel-sized whole or broken shell.

3.2 CURRENT LOGGING TERMS

Definition of terms used on boring logs dated since 2000:

Banded - 0.02' (6 mm) to 0.1' (3.0 cm).
Boulder-Sized – Particles greater than 12 inches in diameter.
Cavity - Voids greater than the diameter of the core.
Coarse Grained - Grain diameter greater than 0.079" (2 mm) for sedimentary rocks or 0.197" (5 mm) for igneous or metamorphic rocks.
Coarse Grained Sand-Sized – Less than 10 percent of fine and medium grained sand sizes are present.
Coarse Gravel-Sized – Particles greater than 3/4 of an inch but less than 3 inches in diameter.
Cobble-Sized – Particles greater than 3 inches but less than 12 inches in diameter.
Decomposed – Applicable to saprolitic rock; rock is essentially reduced to a soil with a relic rock texture; can be molded or crumbled by hand.
Dipping (Dip) - 20 to 45 degrees.
Discontinue – Particles were present within the unit above but are no longer present.
Fine Grained - Grain diameter between 0.004" (0.1 mm) and 0.016" (0.4 mm) for sedimentary rocks or 0.039" (1 mm) for igneous or metamorphic rocks.
Fine Grained Sand-Sized - Less than 10 percent of medium and coarse grained sand sizes are present.
Fine Gravel-Sized – Particles greater than No. 4 sieve but less than 3/4 of an inch in diameter.
Flat (Dip) - 0 to 20 degrees.
Fossiliferous – Greater than 40 percent fossils.
Hard (Hardness) - Difficult to scratch with a knife (cannot be pitted with a geology hammer, but can be chipped with moderate blows of the hammer).
Highly Fractured - Spacing 0.3' (9.1 cm) to 1' (30.5 cm).
Highly Weathered - Entire section is discolored; alteration is greater than 50%; some areas of slightly weathered rock are present; some minerals are leached away; retains only a fraction of its original strength (wet strength usually lower than dry strength).
Intact – Spacing greater than 6' (1.8 m).
Intensely Fractured - Spacing less than 0.3' (9.1 cm).
Massive – Over 3' (0.9 m) thick.
Medium-Bedded - 0.3' (9.1 cm) to 1' (30.5 cm) thick.
Medium Grained - Grain diameters between 0.016" (0.4 mm) to 0.079" (2 mm) for sedimentary rocks or 0.039" (1 mm) to 0.197" (5 mm) for igneous or metamorphic rocks.
Medium Grained Sand-Sized - Less than 10 percent of fine and coarse grained sand sizes are present.
Moderately Fractured - Spacing 1' (30.5 cm) to 3' (0.9 m).
Moderately Hard (Hardness) - Can be scratched easily with a knife, but cannot be scratched by a fingernail (can be pitted with moderate blows of a geology hammer).
Moderately Open (Fracture Aperture) - 0.020" (0.5 mm) to 0.098" (2.5 mm).

Moderately Weathered - Discoloration is evident; surface is pitted and altered, with alterations penetrating well below rock surfaces; 10% to 50% of the rock is altered; strength is noticeably less than unweathered rock.
Non-fossiliferous – No observed fossils.
Open (Fracture Aperture) - 0.098" (2.5 mm) to 0.394" (10 mm).
Pitted - Voids 0.039" (1mm) to 0.236" (6mm) in diameter.
Porous - Voids less than 0.039" (1mm) in diameter.
Slightly Fractured - Spacing 3' (0.9 m) to 6' (1.8 m).
Slightly Weathered - Superficial discoloration, alteration and/or discoloration along discontinuities; less than 10% of the rock volume is altered; strength is essentially unaffected.
Soft (Hardness) - Can be scratched with a fingernail (cannot be crumbled between fingers, but can be easily pitted with light blows of a geology hammer).
Solid - Absence of voids.
Sparsely Fossiliferous – Less than 40 percent fossils.
Steeply Dipping (Dip) - 45 to 90 degrees.
Thick-Bedded – 1' (30.5 cm) to 3' (0.9 m) thick.
Thin-Bedded - 0.1' (3.0 cm) to 0.3' (9.1 cm) thick.
Thin Parting - Paper thin to 0.002' (0.6 mm).
Tight (Fracture Aperture) - 0.004" (0.1 mm) to 0.020" (0.5 mm).
Unweathered - No evidence of any mechanical or chemical alteration.
Very Fine Grained - Grain diameter less than 0.004" (0.1 mm); individual grains or crystals are too small to be seen with the naked eye.
Very Hard (Hardness) - Cannot be scratched with a knife (chips can be broken off only with heavy blows of a geology hammer).
Very Soft (Hardness) - Can be deformed by hand (has a rock-like character, but can be easily broken by hand).
Very Tight (Fracture Aperture) - less than 0.004" (0.1 mm).
Very Wide (Fracture Aperture) - 0.394" (10 mm) to 0.984" (25 mm).
Vuggy - Voids 0.236" (6mm) to the diameter of the core.

3.3 PREVIOUS LOGGING TERMS

Definition of terms used on boring logs dated before 2000:

Dense – Equivalent to SPT N-value of 30 to 50.
Incompetent – Rock that disintegrates while coring; weak.
Indurated – Rock or soil hardened or consolidated by pressure or cementation. Very difficult to break by hand.
Poorly-Indurated – See semi-indurated.
Seam – Rock or soil with average thickness of 2 to 3 inches.
Semi-Indurated – Rock or soil with a lesser degree of hardening or consolidation by pressure or cementation. Crumbles with little effort by hand.

3.4 TESTING AND PROCEDURE METHODS

Test/Procedure	Method
Abrasion Resistance of Large-Size Coarse Aggregate, Los Angeles Machine	ASTM C535
Abrasion Resistance of Small-Size Coarse Aggregate, Los Angeles Machine	ASTM C131
Air Content by the Pressure Method	ASTM C231
Air Content by the Volumetric Method	ASTM C173
Air-Entraining Admixtures for Concrete	ASTM C233
Atterberg Limits, wet preparation method, test method a multipoint	ASTM D4318
Bulk Specific Gravity of Bituminous Mixtures	ASTM D2726

Test/Procedure	Method
Capping Cylindrical Concrete Specimens	ASTM C617
Carbonate Content	ASTM D4373
Compaction: 4" Mold, Modified Proctor	ASTM D1557
Compaction: 4" Mold, Standard Proctor	ASTM D698
Compaction: 6" Mold, Modified Proctor	ASTM D1557
Compaction: 6" Mold, Standard Proctor	ASTM D698
Compressive Strength of Cast in Place Concrete Cylinders	ASTM C873
Compressive Strength of Cylindrical Concrete Specimens (Set of 4)	ASTM C39
Compressive Strength of Lightweight Concrete	ASTM C495
Consolidation, each load of rebound increment with complete time	ASTM D2435
Density and Water Content of Soil by Nuclear Methods (Minimum 4)	ASTM D2922/D3017
Density of Bituminous Concrete in Place by Nuclear Methods (Minimum 4)	ASTM D2950
Density of Soil in Place by the Sand-Cone Method	ASTM D1556
Direct Shear	ASTM D3080
Drilled Cores and Sawed Beams of Concrete	ASTM C42
Extraction of Bituminous Paving Mixtures	ASTM D2172
Freezing and Thawing (up to 55 cycles)	ASTM D5312
Grain size sieve Analysis	ASTM D422
Hydrometer Analysis	ASTM D422
Making and Curing Concrete Test Specimens in the Field (Set of 6)	ASTM C31
Making and Curing Concrete Test Specimens in the Lab (Set of 4)	ASTM C192
Marshall Resistance to Plastic Flow (Set of 3)	ASTM D1559
Material Passing No. 200 Sieve	ASTM D1140
Modulus of Elasticity	ASTM D3148
Moisture Content	ASTM D2216
Munsell Color	Munsell Soil Color Charts
Organic Content, Test Method C	ASTM D2974
Organic Impurities in Fine Aggregate	ASTM C40
Permeability: Constant Head Permeability	ASTM D2434
Permeability: Falling Head Permeability	ASTM D5084
Petrographic Examination	ASTM C295
Sampling Aggregate (Minimum 4 hours)	ASTM D75
Sampling Freshly Mixed Concrete	ASTM C172
Sedimentation Rate	No ASTM
Sieve Analysis of Aggregate	ASTM C136
Slump of Hydraulic-Cement Concrete	ASTM C143
Soil Classification (Boring logs)	ASTM D 2488
Specific Gravity and Absorption of Fine Aggregate	ASTM C128
Specific Gravity for Rock	ASTM C127
Specific Gravity for Soil	ASTM D854
SPLITTING TENSILE STRENGTH	ASTM D3967
Splitting Tensile Strength of Concrete Cylinders	ASTM C496
Subsample Preparation ASTM D4220, Group B	ASTM D4220,
Sulfate Soundness	ASTM C88

Test/Procedure	Method
Total Evaporable Moisture Content	ASTM C566
Triaxial Compression Test for Rock with Strain Gages or LVDTs (per confining pressure)	ASTM D2664
Triaxial compression: Consolidated Undrained (CU) Triaxial Compression Test R Test with Pore Pressure Measurements (per confining pressure)	ASTM D4767
Triaxial compression: Unconsolidated Undrained (UU) Triaxial Compression Test Q Test with Pore Pressure Measurements (per confining pressure)	ASTM D2850
Unbonded Caps for Compressive Strength	ASTM C1231
Unconfined Compressive Strength for Rock Greater than 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Rock up to 9,000 psi with Strain	ASTM D2938
Unconfined Compressive Strength for Soil	ASTM D2166
Unit Weight and Absorption	ASTM C127
Unit Weight of Aggregate	ASTM C29 -
Visual Percent Shell	No ASTM
Wetting and Drying	ASTM D5313

4 GEOTECHNICAL DATA

4.1 SUMMARY OF FIELD INVESTIGATIONS

The tables below summarizes the field investigations data set available for this project.

Table 1. Available Vibracore Data

Designation	State Plane, FL-East, NAD83		Project Location
	X	Y	
VH-FH18-1	509586	2307443	Turning Basin
VH-FH18-2	509202	2306561	
* Coordinates presented correspond to the project coordinate system and datum			

Table 2. Available Surficial Samples Data

Designation	State Plane, FL-East, NAD83		Project Location
	X	Y	
SS-FH18-1	511222	2309132	Cut 5
* Coordinates presented correspond to the project coordinate system and datum			

4.2 SUMMARY OF INDEX TESTING DATA

The table below summarizes the index testing available for this project.

Table 3: Index Testing Available for this Project

Designation	Depth (ft)	Sample Number	USCS	LL	PL	PI	Visual Shell %	Munsell Color
SS-FH18-1	0	N/A	SP	NP	NV	NP	42.2	5Y 3/1
VB-FH18-1	4	N/A1	SC	77	25	52		5Y 2.5/1
	10	N/A2	SC					10Y 2.5/1
	13	N/A3	CH	169	41	128		10Y 3/1
	19	N/A4	SP-SM				0	5Y 4/1
VB-FH18-2	4	N/A1	SM				12.7	5Y 3/1
	9	N/A2	OH	81	28	53		5Y 2.5/1
	10.5	N/A3	SM				44.1	5Y 3/1
	16	N/A4	SP				0	2.5Y 7/1

4.3 BORING LOG NOTES

Borings VB-FH18-1 and VB-FH18-2 were sampled using an Alpine style, pneumatic vertical hammer, vibracore drilling apparatus having a continuous 20-foot by 3 or 3 1/2-inch core barrel with clear plastic core liner that was advanced until a depth of 20 feet was reached or until refusal was encountered. Refusal is defined as no additional penetration of the vibracore with two minutes of additional effort applied to penetrate through the resistant layer at maximum vibration.

Surface Sample SS-FH18-1 was collected with an AMS 6”x6”x6” clam shell sampler which weighs approximately 25 pounds. The sampler was tied to nylon rope and mounted on a shipboard winch with pulley. During sampling, the clam shell was allowed to free-fall from the winch at the target location.

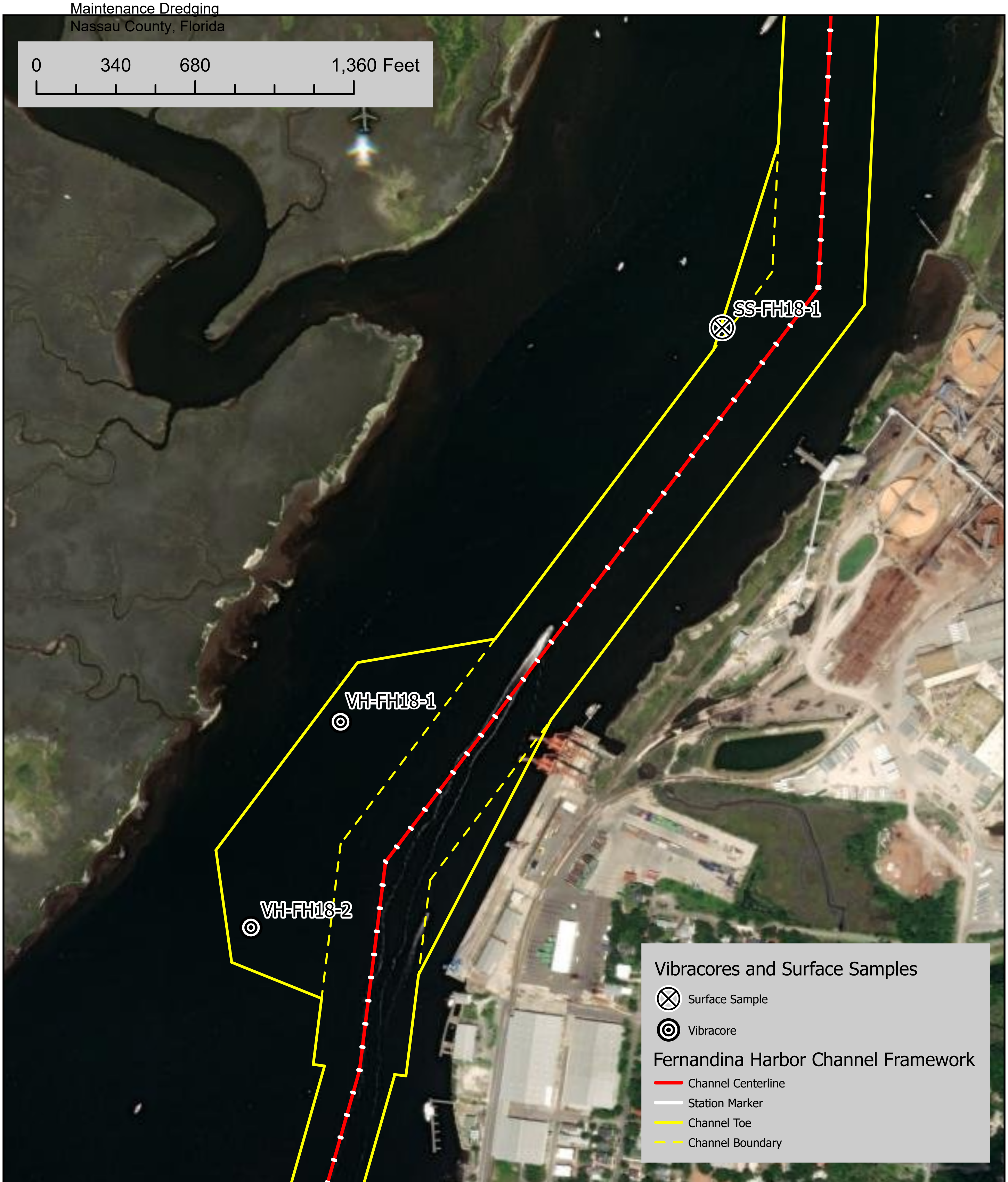
4.4 INSPECTION OF RECOVERED MATERIALS

The core borings for this project are not available for inspection.

US Army Corps of Engineers The Department of the Army Jacksonville District	Fernandina Harbor Maintenance Dredging Nassau County, Florida	N
Date: 2/11/2025	Dsn by: NRM · Drwn by: NRM · Ckd by: JLC	PLATE NO. 1

This map is not intended for the use of construction.

Fernandina Harbor
Maintenance Dredging
Nassau County, Florida



US Army Corps of Engineers
The Department of the Army
Jacksonville District

Date: 3/10/2025

Fernandina Harbor
Maintenance Dredging
Nassau County, Florida

N

PLATE NO. 2

Dsn by: NRM | Drwn by: NRM | Ckd by: JLC

This map is not intended for the use of construction.

4.5 BORING LOGS

Applicable boring logs are presented on the following pages.

While the Government's borings are representative of subsurface conditions at their respective locations and vertical reaches, local variations in the characteristics of the subsurface materials of this region are to be expected. Accordingly, prospective offerors shall form their own conclusions from the examination of the recovered materials prior to submission of their offer.

Boring Designation VH-FH18-1

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: Fernandina Harbor O&M	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, FLE (U.S. Ft.) — HORIZONTAL: NAD83 — VERTICAL: MLLW
2. BORING DESIGNATION: VH-FH18-1 — LOCATION COORDINATES: X = 509,586 Y = 2,307,444	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ — CONTRACTOR FILE NO.:	12. TOTAL SAMPLES — DISTURBED: 4 — UNDISTURBED (UD): 0
4. NAME OF DRILLER: SAW Snell	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED — DEG. FROM VERTICAL: — BEARING:	14. ELEVATION GROUND WATER:
	15. DATE BORING — STARTED: 08-19-18 — COMPLETED: 08-19-18
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -28.2 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 100 %
8. TOTAL DEPTH OF BORING: 20.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Scott Davidson, Geologist

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-28.2	0.0						-28.2		
			SILT, organic-H, some silt, little fine-grained sand-sized quartz, trace shell, weak reaction with HCl, 5Y 2.5/1 black (OH)	100			Vibracore		
-32.2	4.0						-32.2		
			SAND, clayey, mostly fine-grained sand-sized quartz, some clay, trace shell, weak reaction with HCl, 5Y 2.5/1 black (SC)	100	N/A1		Vibracore		
							-38.2		
			From El. -38.2 to -39.2 Ft., mostly fine-grained sand-sized quartz, little clay, trace organic matter, weak reaction with HCl, 10Y 2.5/1 greenish black	100	N/A2		Vibracore		
-39.2	11.0						-39.2		
			CLAY, fat, mostly clay, trace quartz, no reaction with HCl, 10Y 3/1 very dark greenish gray (CH)	100			Vibracore		
							-41.2		
				100	N/A3		Vibracore		

0
5
10
15

SAJ FORM 1836
JUN 02

(Continued)

Boring Designation VH-FH18-1

DRILLING LOG (Cont. Sheet)	INSTALLATION Jacksonville District		SHEET 2 OF 2 SHEETS
PROJECT Fernandina Harbor O&M	COORDINATE SYSTEM/DATUM State Plane, FLE (U.S. Ft.)	HORIZONTAL NAD83	VERTICAL MLLW
LOCATION COORDINATES X = 509,586 Y = 2,307,444	ELEVATION TOP OF BORING -28.2 Ft.		

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-46.7	18.5			100			Vibracore -46.7		
			SAND, poorly-graded with silt, mostly fine to medium-grained sand-sized quartz, few silt, no reaction with HCl, 5Y 4/1 dark gray (SP-SM)	100			Vibracore -47.2		
-48.2	20.0			100	N/A4		Vibracore -48.2		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
N/A1	4.0/4.5	SC
N/A2	10.0/10.5	SC*
N/A4	19.0/19.5	SP-SM*

*Lab visual classification based on gradation curve

SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT
4.0	Atterberg (PI)	52	
13.0	Atterberg (PI)	128	

15 20 25 30 35

Fernandina Harbor
Maintenance Dredging
Nassau County, Florida

Boring Designation VH-FH18-2

DRILLING LOG	DIVISION: South Atlantic	INSTALLATION: Jacksonville District	SHEET 1 OF 2 SHEETS

1. PROJECT: Fernandina Harbor O&M	9. SIZE AND TYPE OF BIT: See Remarks
	10. COORDINATE SYSTEM/DATUM: State Plane, FLE (U.S. Ft.) — HORIZONTAL: NAD83 — VERTICAL: MLLW
2. BORING DESIGNATION: VH-FH18-2 — LOCATION COORDINATES: X = 509,203 Y = 2,306,561	11. MANUFACTURER'S DESIGNATION OF DRILL: ☐ AUTO HAMMER ☐ MANUAL HAMMER
3. DRILLING AGENCY: Corps of Engineers - CESAJ — CONTRACTOR FILE NO.:	12. TOTAL SAMPLES — DISTURBED: 4 — UNDISTURBED (UD): 0
4. NAME OF DRILLER: SAW Snell	13. TOTAL NUMBER CORE BOXES: 0
5. DIRECTION OF BORING: ☒ VERTICAL ☐ INCLINED — DEG. FROM VERTICAL: — BEARING:	14. ELEVATION GROUND WATER:
	15. DATE BORING — STARTED: 08-19-18 — COMPLETED: 08-19-18
6. THICKNESS OF OVERBURDEN: N/A	16. ELEVATION TOP OF BORING: -27.2 Ft.
7. DEPTH DRILLED INTO ROCK: N/A	17. TOTAL RECOVERY FOR BORING: 100 %
8. TOTAL DEPTH OF BORING: 20.0 Ft.	18. SIGNATURE AND TITLE OF INSPECTOR: Scott Davidson, Geologist

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
-27.2	0.0						-27.2		
-29.2	2.0		SILT, organic-H, mostly silt, trace shell, no reaction with HCl, 5Y 2.5/1 black (OH)	100			Vibracore -29.2		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, little fine-grained sand-sized shell, few organic matter, weak reaction with HCl, 5Y 3/1 very dark gray (SM)	100			Vibracore -31.2		
-34.2	7.0			100	N/A1		Vibracore -34.2		
			SILT, organic-H, mostly silt, little fine-grained sand-sized quartz, trace shell, weak reaction with HCl, 5Y 2.5/1 black (OH)	100			Vibracore -36.2		
-37.2	10.0			100	N/A2		Vibracore -37.2		
			SAND, silty, mostly fine-grained sand-sized quartz, little silt, few organic matter, trace shell, strong reaction with HCl, 5Y 3/1 very dark gray (SM)	100			Vibracore -37.7		
-38.7	11.5			100	N/A3		Vibracore -38.7		
-40.6	13.4		SAND, clayey, some fine-grained sand-sized quartz, little clay, little plant debris, few organic matter, weak reaction with HCl, 10YR 2/2 very dark brown (SC)	100			Vibracore -40.6		
			SAND, poorly-graded, mostly fine-grained sand-sized quartz, little plant debris, no reaction with HCl, 2.5Y 7/1 light gray (SP)	100			Vibracore		

SAJ FORM 1836
JUN 02

(Continued)

Boring Designation VH-FH18-2

DRILLING LOG (Cont. Sheet)	INSTALLATION: Jacksonville District			SHEET 2 OF 2 SHEETS
PROJECT: Fernandina Harbor O&M	COORDINATE SYSTEM/DATUM: State Plane, FLE (U.S. Ft.)	HORIZONTAL: NAD83	VERTICAL: MLLW	
LOCATION COORDINATES: X = 509,203 Y = 2,306,561	ELEVATION TOP OF BORING: -27.2 Ft.			

ELEV. (ft)	DEPTH (ft)	LEGEND	CLASSIFICATION OF MATERIALS	% REC.	BOX OR SAMPLE	RQD OR UD	REMARKS	BLOWS/ 1 FT.	N-VALUE
				100			Vibracore -43.2		
					N/A4				
				100			Vibracore -45.9		
			From El. -45.9 to -46.6 Ft., mostly fine to medium-grained sand-sized quartz, no reaction with HCl, 2.5Y 6/2 light brownish gray	100			Vibracore -46.6		
-47.2	20.0		From El. -46.6 to -47.2 Ft., mostly fine to medium-grained sand-sized quartz, little plant debris, no reaction with HCl, 2.5Y 6/2 light brownish gray	100			Vibracore -47.2		

NOTES:

1. USACE Jacksonville is the custodian for these original files.

2. Soils are field visually classified in accordance with the Unified Soils Classification System.

3. Laboratory Testing Results

SAMPLE ID	SAMPLE DEPTH	LABORATORY CLASSIFICATION
N/A1	4.0/4.5	SM*
N/A2	9.0/9.5	OH
N/A3	10.5/11.0	SM*
N/A4	16.0/16.5	SP*

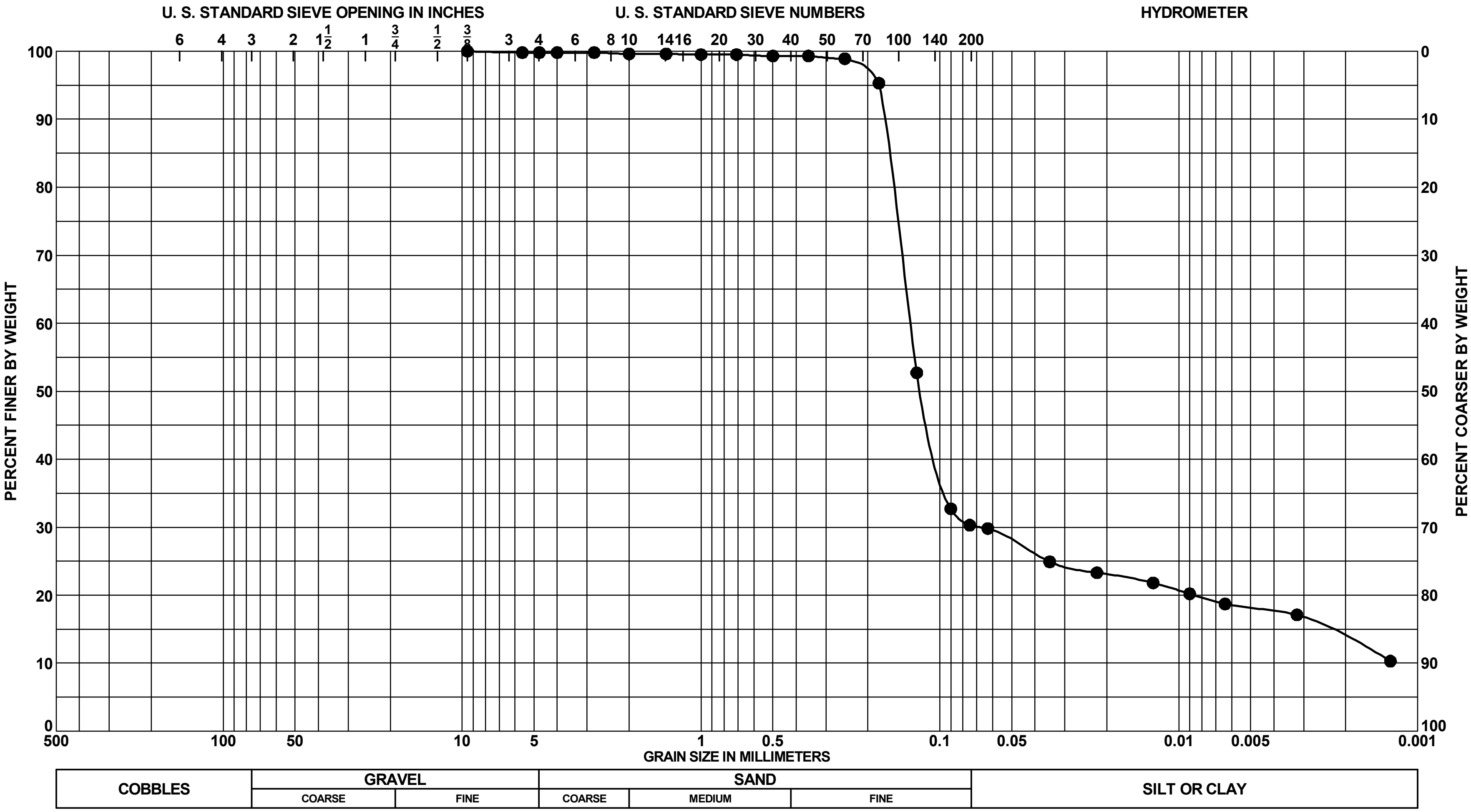
*Lab visual classification based on gradation curve

SAMPLE DEPTH	LABORATORY SOIL TESTING	RESULT	UNIT
9.0	Atterberg (PI)	53	

15 20 25 30 35

4.6 LABORATORY TESTING DATA

Applicable laboratory testing data are presented on the following pages.

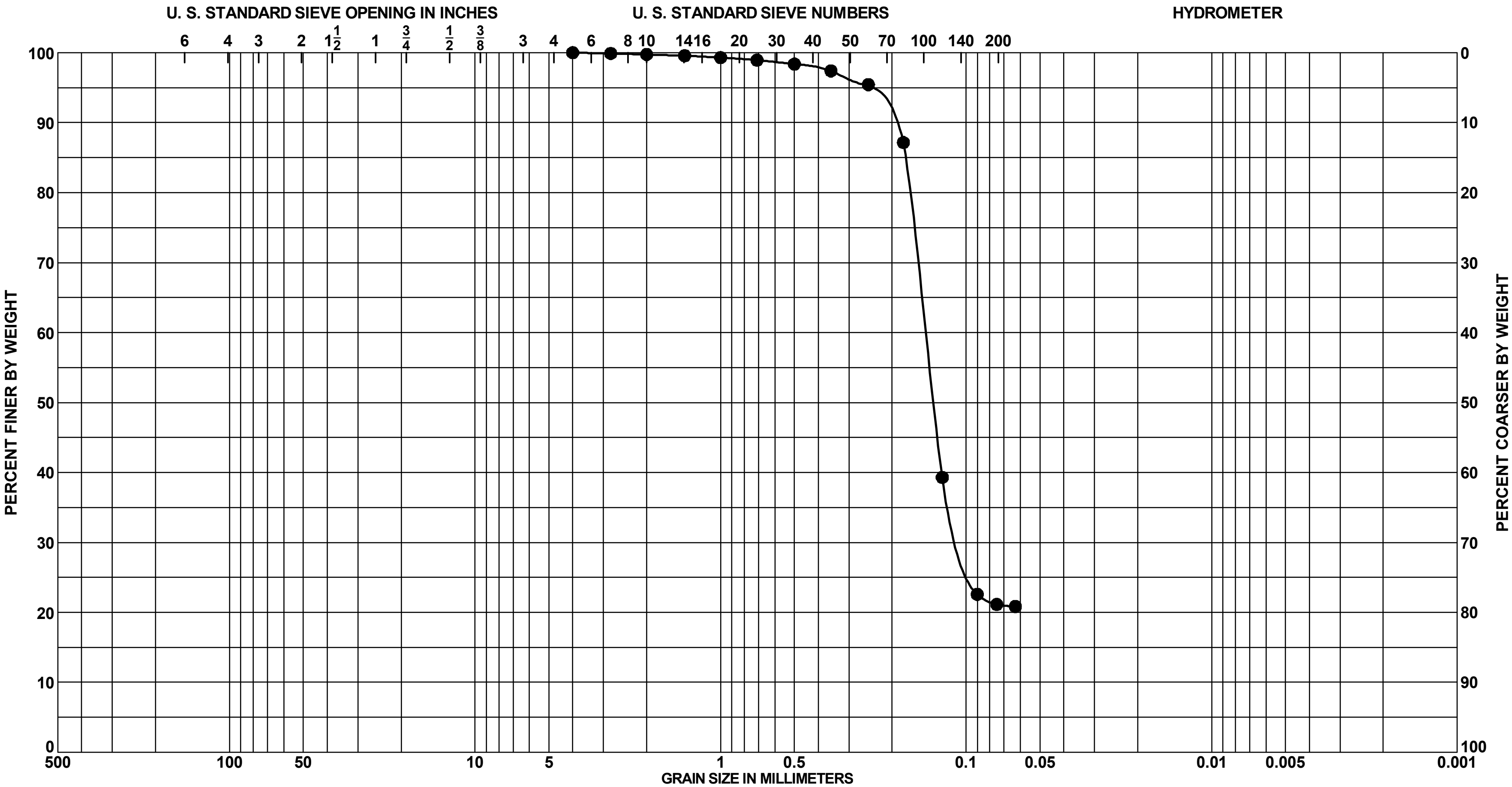


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	4.0 to 4.5 Ft.	SAND, clayey, mostly fine-grained sand-sized quartz, some clay, trace shell, black (SC)	5Y 2.5/1						77	25	52

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-1
BORING ELEV.	-28.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02



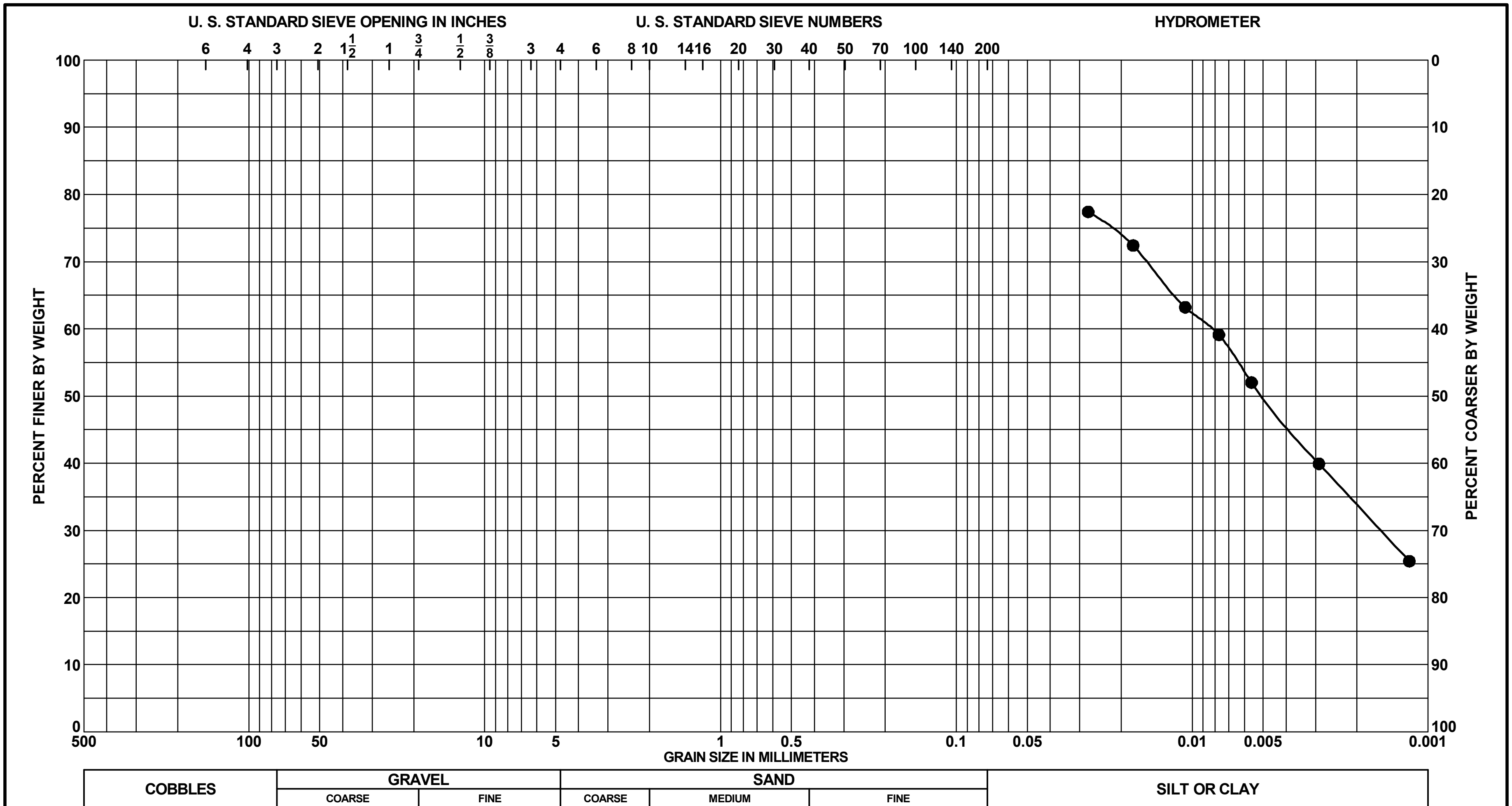
COBBLES	GRAVEL		SAND			SILT OR CLAY
	COARSE	FINE	COARSE	MEDIUM	FINE	

Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	10.0 to 10.5 Ft.	SAND, clayey,mostly fine-grained sand-sized quartz, little clay, trace organic matter, greenish black (SC)	10Y 2.5/1								

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-1
BORING ELEV.	-28.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02

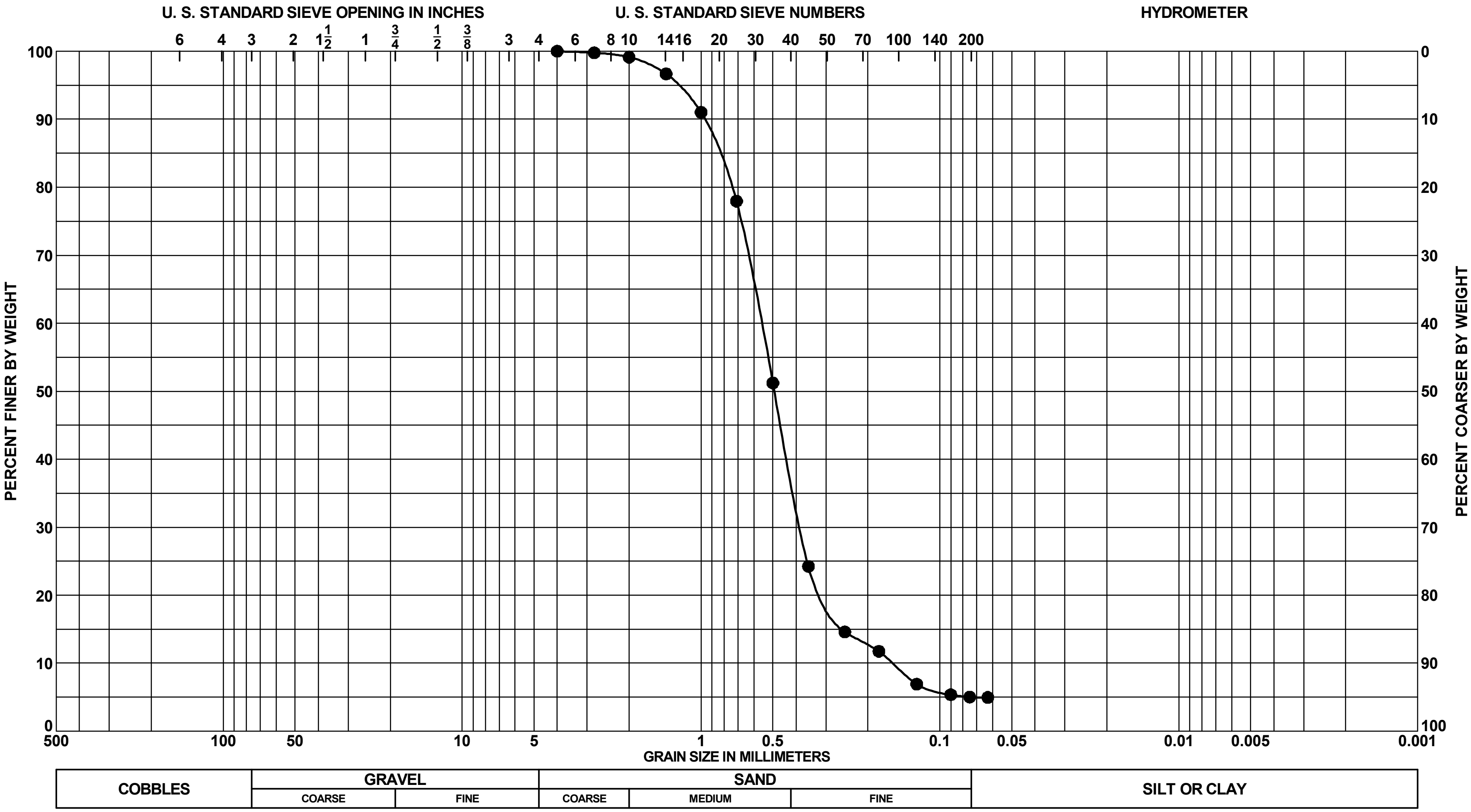


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	13.0 to 13.5 Ft.	CLAY, inorganic-H, mostly clay, trace	10Y 3/1						169	41	128
		quartz, very dark greenish gray (CH)									

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-1
BORING ELEV.	-28.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02



Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● N/A	19.0 to 19.5 Ft.	SAND, poorly-graded with silt, mostly	5Y 4/1	0							
		fine to medium-grained sand-sized									
		quartz, few silt, dark gray (SP-SM)									

PROJECT King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH

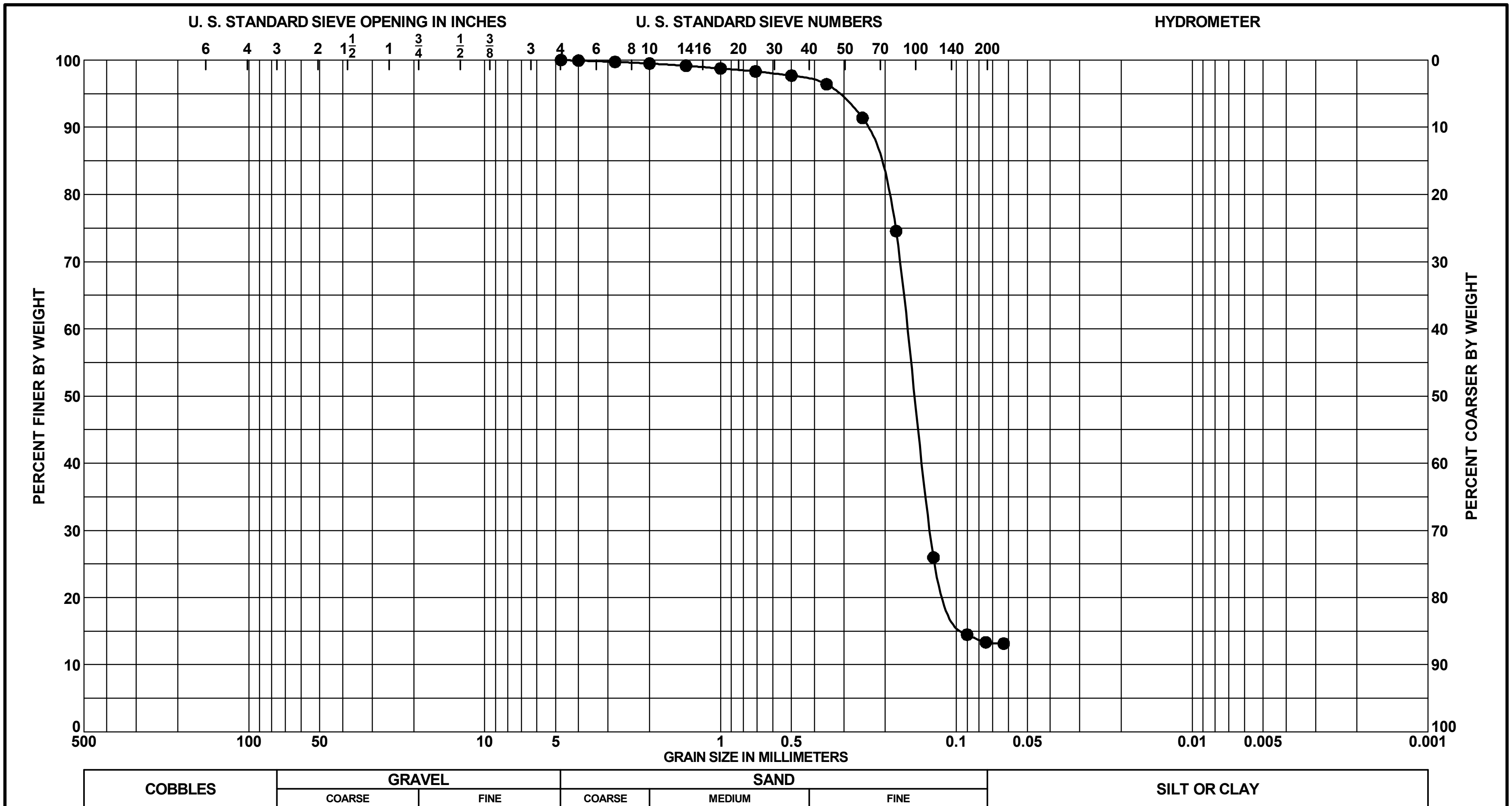
BORING NO. VB-FH18-1

BORING ELEV. -28.2 Ft., MLLW

DATE 11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02

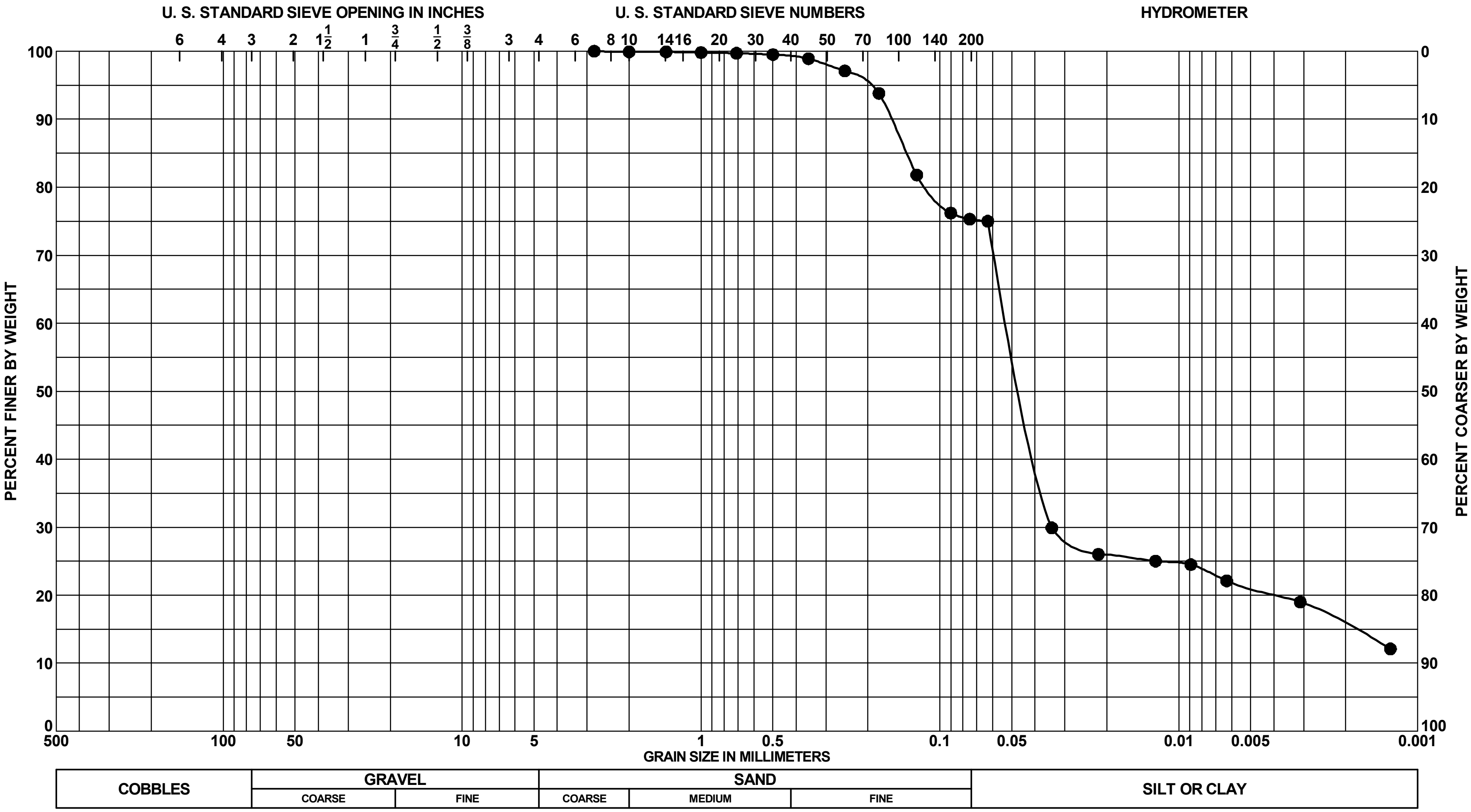


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	4.0 to 4.5 Ft.	SAND, silty, mostly fine-grained	5Y 3/1	13							
		sand-sized quartz, little silt, few									
		organic matter, little fine-grained									
		sand-sized shell, very dark gray (SM)									

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-2
BORING ELEV.	-27.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02

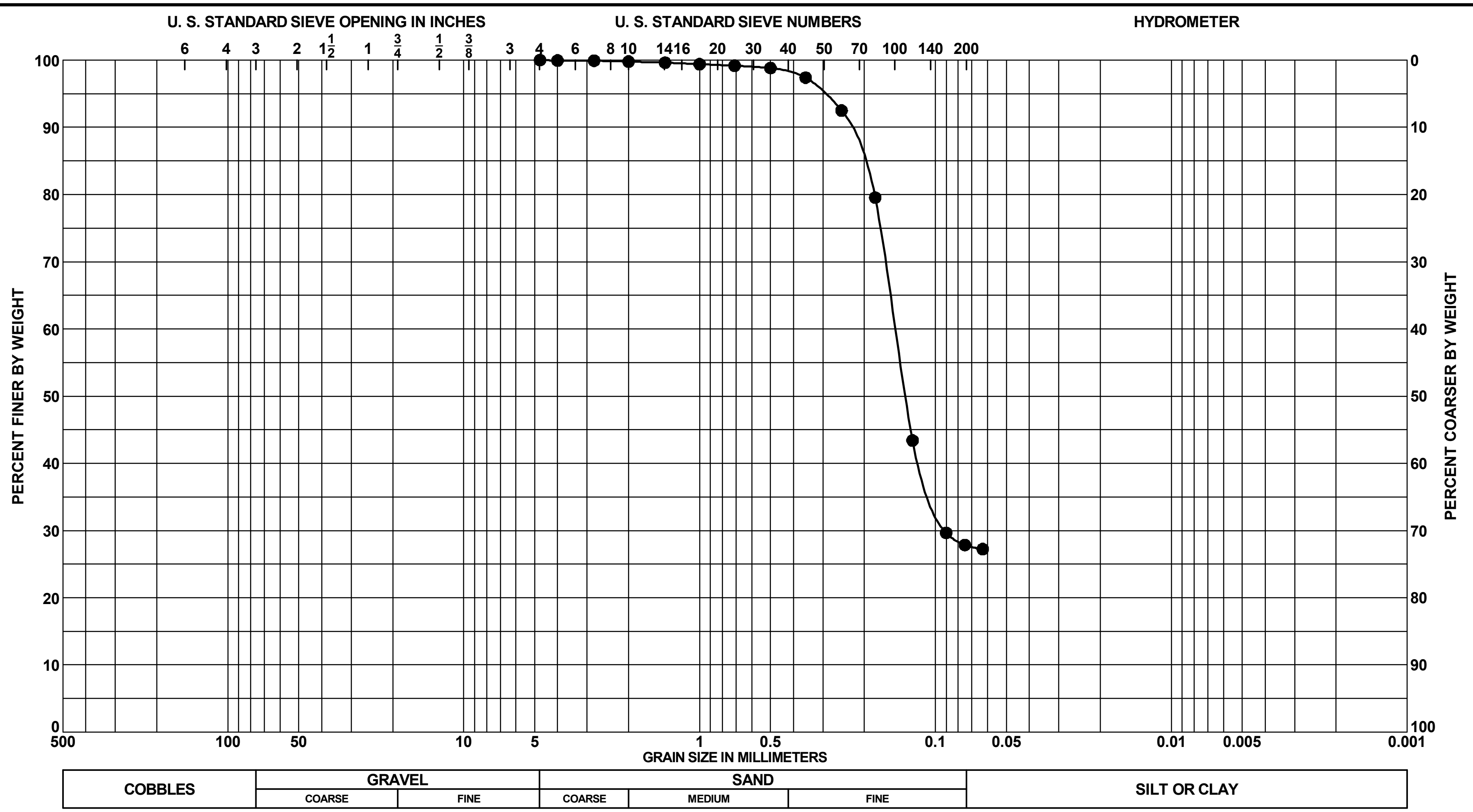


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	9.0 to 9.5 Ft.	SILT, organic-H, mostly silt, little fine-grained sand-sized quartz, trace shell, black (OH)	5Y 2.5/1						81	28	53

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-2
BORING ELEV.	-27.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02

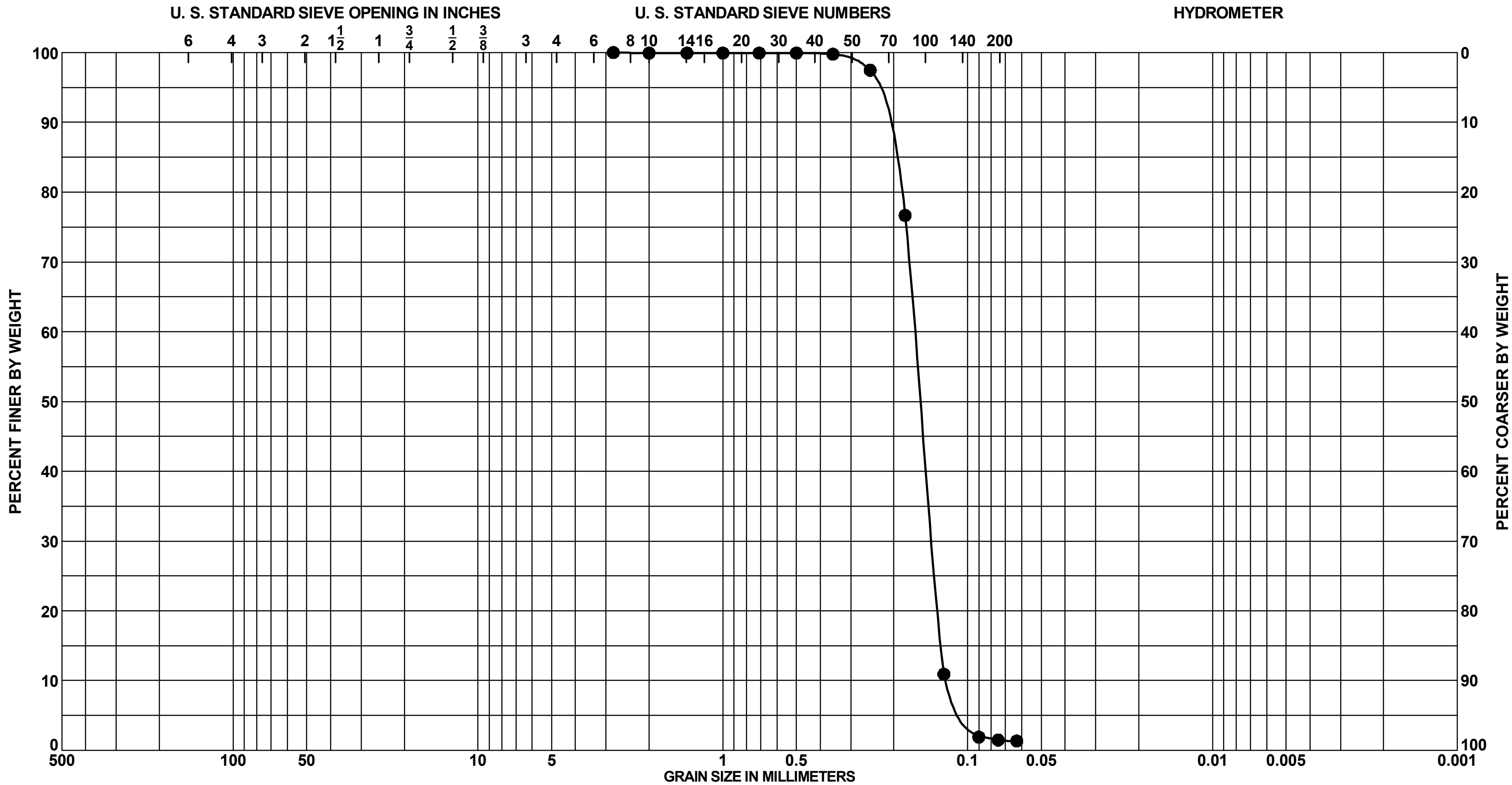


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	$\%CO_3$	G_s	Org %	w_n	LL	PL	PI
● N/A	10.5 to 11.0 Ft.	SAND, silty, mostly fine-grained sand-sized quartz, little silt, few organic matter, trace shell, very dark gray (SM)	5Y 3/1	4							

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-2
BORING ELEV.	-27.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02

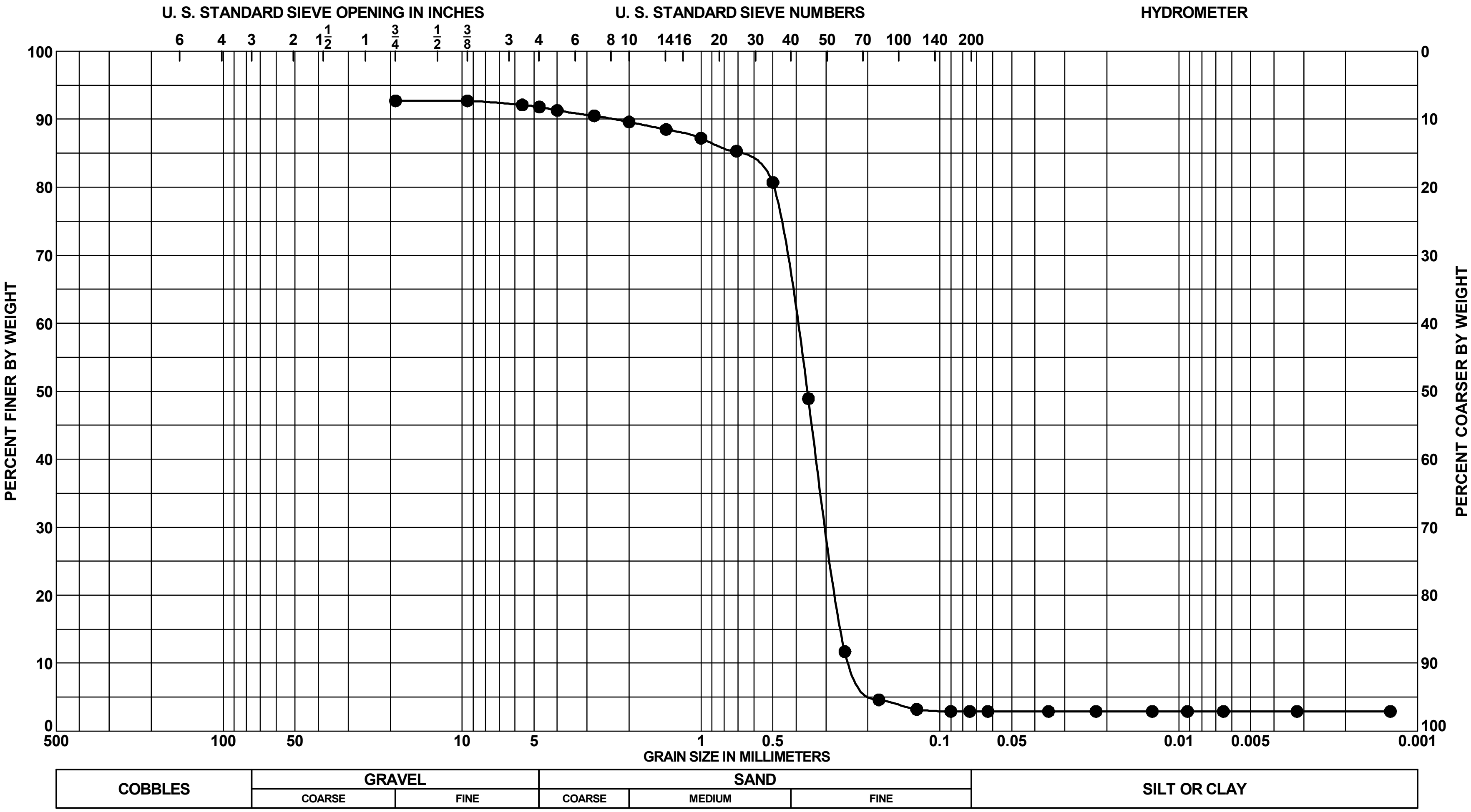


Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	16.0 to 16.5 Ft.	SAND, poorly-graded, mostly	2.5Y 7/1	0							
		fine-grained sand-sized quartz, little									
		plant debris, light gray (SP)									

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	VB-FH18-2
BORING ELEV.	-27.2 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02



Sample No.	Depth	Classification	Munsell	Visual % Shell (est)	%CO_3	G_s	Org %	w_n	LL	PL	PI
● N/A	0.0 to 1.0 Ft.	SAND, poorly-graded, mostly fine to medium-grained sand-sized quartz, some fine-grained sand-sized to coarse gravel-sized shell up to 4", trace silt, very dark gray (SP)	5Y 3/1	42					NP	NV	NP

PROJECT	King's Bay Entrance Channel and Fernandina Harbor - North Settling Basin and FH
BORING NO.	SS-FH18-1
BORING ELEV.	-36.6 Ft., MLLW
DATE	11/2/2018

GRADATION CURVES

SAJ FORM 2087
JUN 02